

Feed Forward Neural Networks in Depth (1)

FFNNs

- contain an input and an output layer with at least one hidden layer in between.
- Fully-connected layers, which means that each node in one layer connects to every node in the following layer.
- using activation functions, one can introduce non linearity ~~in the net~~.

~~Forward Propagation~~

Mathematics behind forward propagation

forward propagation: involves computing predictions and the associated cost of these predictions.

These will be computed:

- multiple layers.
- several nodes in each layer.
- various activation functions.
- various types of cost functions.
- mini batches of training examples.

- mathematical ~~not~~ calculations of anode:

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$$z_{j,i}^{[l]} = \sum_k w_{j,k}^{[l]} \cdot a_{k,i}^{[l-1]} + b_j^{[l]},$$

$$a_{j,i}^{[l]} = g_j(z_{1,i}^{[l]}, \dots, z_{j,i}^{[l]}, z_{n^{[l]},i}^{[l]}).$$

l : the current layer ($l = 1, \dots, L$), where L is the number of layers that have weights and biases (means hidden layers and output layer).

so $b^L = 0$ in output layer $l = L$ output layer.

$n^{[l]}$: the number of nodes in the current layer.
 $1, \dots, n^{[l]}$

$n^{[l-1]}$: the number of nodes in the previous layer
 $1, \dots, n^{[l-1]}$

i : the current training example $i = 1, \dots, m$, where m is the number of training examples.

$z_{j,i}^{[l]}$ = a weighted sum of the activations of the previous layer, shifted by a bias.

$w_{j,k}^{[l]}$ = a weight that scales the k th activation of the previous layer.

③

$b_j^{[L]}$ = a bias in the current layer.

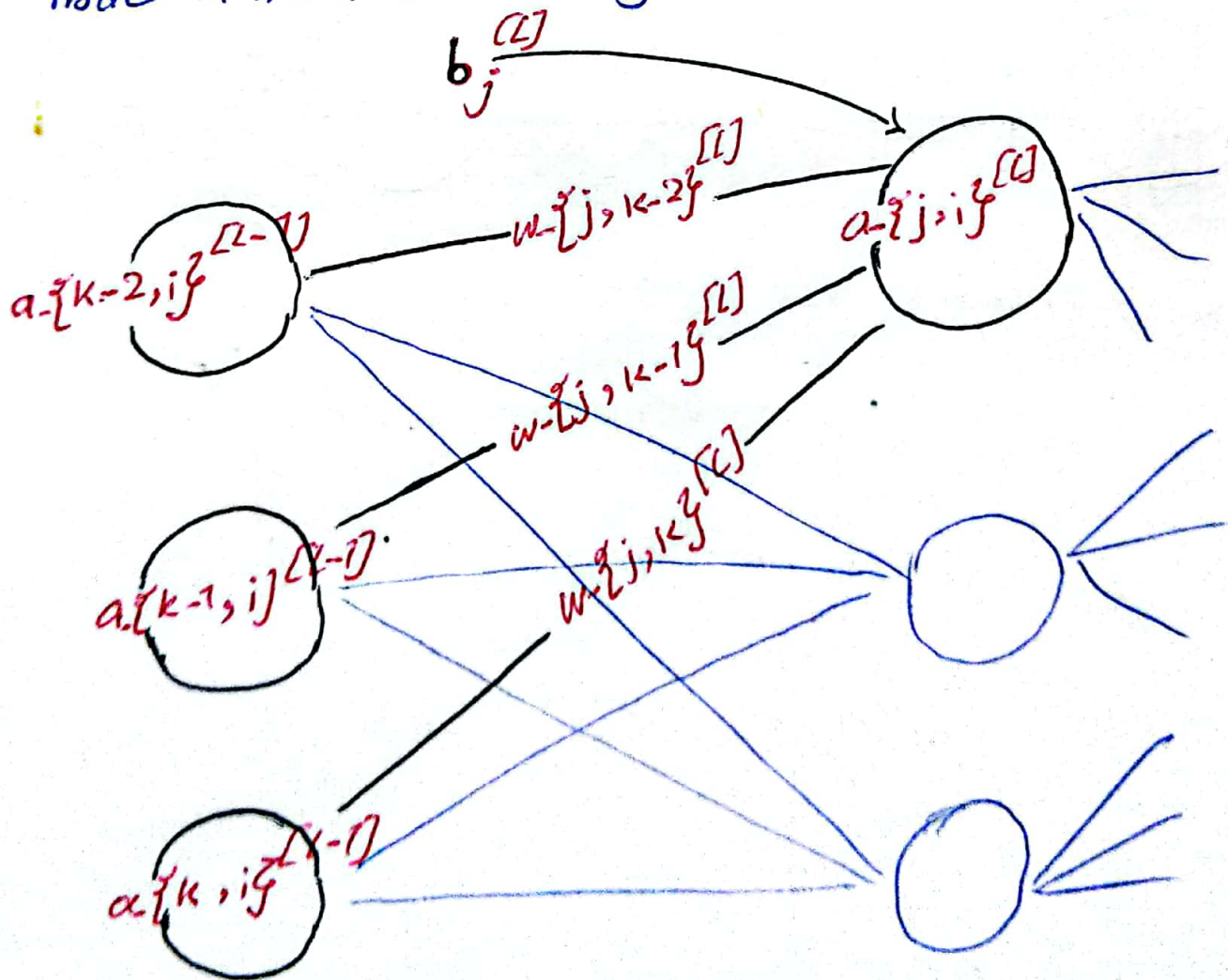
$a_{j,i}^{[L]}$ = an activation in the current layer.

$a_{k,i}^{[L-1]}$: an activation in the previous layer.

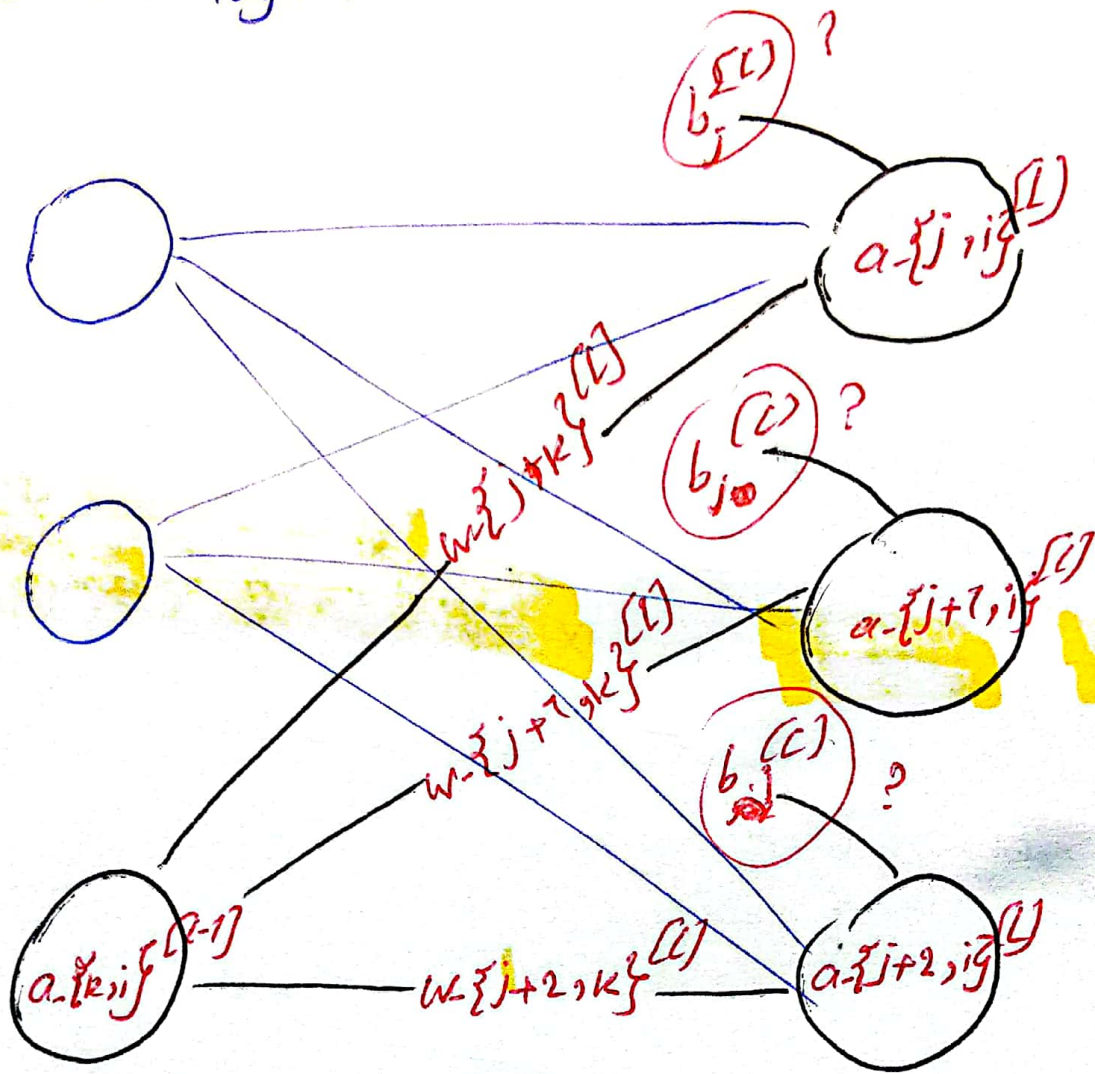
$g_j^{[L]}$: an activation function $g_j^{[L]}: \mathbb{R}^{n^{[L]}} \rightarrow \mathbb{R}$ used in the current layer.

$A^{[0]} = X$; $A^{[L]} = \hat{Y} \Rightarrow X \in \mathbb{R}^{n \times m}$ & $\hat{Y} \in \mathbb{R}^{n, m}$

- A node in the current layer depends on every node in the previous layer:



- A node in the previous layer affects every node in the current layer.



vectorization the nodes:

$$\begin{bmatrix} z_{1,i}^{(l)} \\ \vdots \\ z_{j,i}^{(l)} \\ \vdots \\ z_{n,i}^{(l)} \end{bmatrix} = \begin{bmatrix} w_{1,1}^{(l)} & \dots & w_{1,k}^{(l)} & \dots & w_{1,n}^{(l-1)} \\ \vdots & & \vdots & & \vdots \\ w_{j,1}^{(l)} & \dots & w_{j,k}^{(l)} & \dots & w_{j,n}^{(l-1)} \\ \vdots & & \vdots & & \vdots \\ w_{n,1}^{(l)} & \dots & w_{n,k}^{(l)} & \dots & w_{n,n}^{(l-1)} \end{bmatrix} \begin{bmatrix} a_{1,i}^{(l-1)} \\ \vdots \\ a_{k,i}^{(l-1)} \\ \vdots \\ a_{n,i}^{(l-1)} \end{bmatrix} + \begin{bmatrix} b_1^{(l)} \\ \vdots \\ b_j^{(l)} \\ \vdots \\ b_n^{(l)} \end{bmatrix}$$

$$\begin{bmatrix} a_{1,i}^{(L)} \\ \vdots \\ a_{j,i}^{(L)} \\ \vdots \\ a_{n,i}^{(L)} \end{bmatrix} = \begin{bmatrix} g_1^{(L)}(z_{1,i}^{(L)}, \dots, z_{j,i}^{(L)}, \dots, z_{n,i}^{(L)}) \\ \vdots \\ g_j^{(L)}(z_{1,i}^{(L)}, \dots, z_{j,i}^{(L)}, \dots, z_{n,i}^{(L)}) \\ \vdots \\ g_{n,i}^{(L)}(z_{1,i}^{(L)}, \dots, z_{j,i}^{(L)}, \dots, z_{n,i}^{(L)}) \end{bmatrix} \quad (5)$$

which we can write as:

$$z_{:,i}^{(L)} = W \cdot a_{:,i}^{(L-1)} + b^{(L)}$$

$$a_{:,i}^{(L)} = g^{(L)}(z_{:,i}^{(L)})$$

Vectorization of the training examples:

$$\begin{aligned} Z^{(L)} &= \begin{bmatrix} z_{:,1}^{(L)} & \dots & z_{:,i}^{(L)} & \dots & z_{:,m}^{(L)} \end{bmatrix} \\ &= W^{(L)} \begin{bmatrix} a_{:,1}^{(L-1)} & \dots & a_{:,i}^{(L-1)} & \dots & a_{:,m}^{(L-1)} \end{bmatrix} + \begin{bmatrix} b^{(L)} & \dots & b^{(L)} & \dots & b^{(L)} \end{bmatrix} \\ &= W^{(L)} \cdot A^{(L-1)} + \text{broadcast}(b^{(L)}), \end{aligned}$$

$$A^{(L)} = \begin{bmatrix} a_{:,1}^{(L)} & \dots & a_{:,i}^{(L)} & \dots & a_{:,m}^{(L)} \end{bmatrix} \quad (?)$$

Cost function:

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$$J = f(\hat{Y}, Y) = f(A^{[1]}, Y)$$

mathematics behind (backward propagation)
Backpropagation

Partial derivative and chain rule:

chain rule (exp):

$$\begin{cases} a_i = g_i(x_1, \dots, x_j, \dots, x_n) \\ y_k = f_k(a_1, \dots, a_i, \dots, a_m) \end{cases}$$

$$\frac{\partial y_k}{\partial x_j} = \sum_i \frac{\partial y_k}{\partial a_i} \cdot \frac{\partial a_i}{\partial x_j}$$

$$\frac{\partial J}{\partial w_{j,k}^{(l)}} = \sum_i \frac{\partial J}{\partial z_{ji}^{(l)}} \cdot \frac{\partial z_{ji}^{(l)}}{\partial w_{j,k}^{(l)}} = \sum_i \frac{\partial J}{\partial z_{ji}^{(l)}} a_{k,i}^{(l-1)} \quad (7)$$

$$\frac{\partial J}{\partial b_j^{(l)}} = \sum_i \frac{\partial J}{\partial z_{ji}^{(l)}} \cdot \frac{\partial z_{ji}^{(l)}}{\partial b_j^{(l)}} = \sum_i \frac{\partial J}{\partial z_{ji}^{(l)}}$$

vectorization results of the equation above:

$$\frac{\partial J}{\partial w^{(l)}} = \sum_i \frac{\partial J}{\partial z_{:,i}^{(l)}} \cdot a_{:,i}^{(l-1)T} = \frac{\partial J}{\partial z^{(l)}} A^{(l-1)T}$$

$$\frac{\partial J}{\partial b^{(l)}} = \sum_i \frac{\partial J}{\partial z_{:,i}^{(l)}} = \underbrace{\sum_{\text{axis}=1}_{\text{column vector}} \frac{\partial J}{\partial z^{(l)}}}$$

where: $\partial J / \partial z_{:,i}^{(l)} \in \mathbb{R}^{n^{(l)}}$, $\partial J / \partial z^{(l)} \in \mathbb{R}^{n^{(l)} \times m}$, $\partial J / \partial w^{(l)} \in \mathbb{R}^{h^{(l)} \times n^{(l-1)}}$
 $\partial J / \partial b^{(l)} \in \mathbb{R}^{n^{(l)}}$

$$\frac{\partial J}{\partial z_{j,i}^{(l)}} = \sum_{p=1}^n \frac{\partial J}{\partial a_{p,i}^{(l)}} \cdot \frac{\partial a_{p,i}^{(l)}}{\partial z_{j,i}^{(l)}} \quad p=1, \dots, n^{(l)}$$

Vectorized version,

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$$\frac{\partial J}{\partial z_{ji}^{[l]}} = \frac{\partial a_{ji}^{[l]}}{\partial z_{ji}^{[l]}} \cdot \frac{\partial J}{\partial a_{ji}^{[l]}}$$

where $\frac{\partial J}{\partial a_{ji}^{[l]}} \in \mathbb{R}^{n^{[l]}}$, $\frac{\partial a_{ji}^{[l]}}{\partial z_{ji}^{[l]}} \in \mathbb{R}^{n^{[l]} \times n^{[l]}}$

$$\frac{\partial J}{\partial \mathbf{z}^{[l]}} = \left[\frac{\partial J}{\partial z_{j1}^{[l]}}, \dots, \frac{\partial J}{\partial z_{ji}^{[l]}}, \dots, \frac{\partial J}{\partial z_{jm}^{[l]}} \right]$$

$$\frac{\partial J}{\partial \mathbf{A}^{[l]}} = \left[\frac{\partial J}{\partial a_{j1}^{[l]}}, \dots, \frac{\partial J}{\partial a_{ji}^{[l]}}, \dots, \frac{\partial J}{\partial a_{jm}^{[l]}} \right]$$

$$\frac{\partial J}{\partial \mathbf{A}^{[l]}} \in \mathbb{R}^{n^{[l]} \times m}$$

$$\frac{\partial J}{\partial a_{kji}^{[l-1]}} = \sum_j \frac{\partial J}{\partial z_{jji}^{[l]}} \cdot \frac{\partial z_{jji}^{[l]}}{\partial a_{kji}^{[l-1]}} = \sum_j \frac{\partial J}{\partial z_{jji}^{[l]}} w_{jik}^{[l]}$$

$$\frac{\partial J}{\partial A^{[L-1]}} = W^{[L]T} \frac{\partial J}{\partial Z^{[L]}}$$

$$\frac{\partial J}{\partial A^{[L-1]}} \in \mathbb{R}^{n^{[L-1]} \times m}$$

Summary

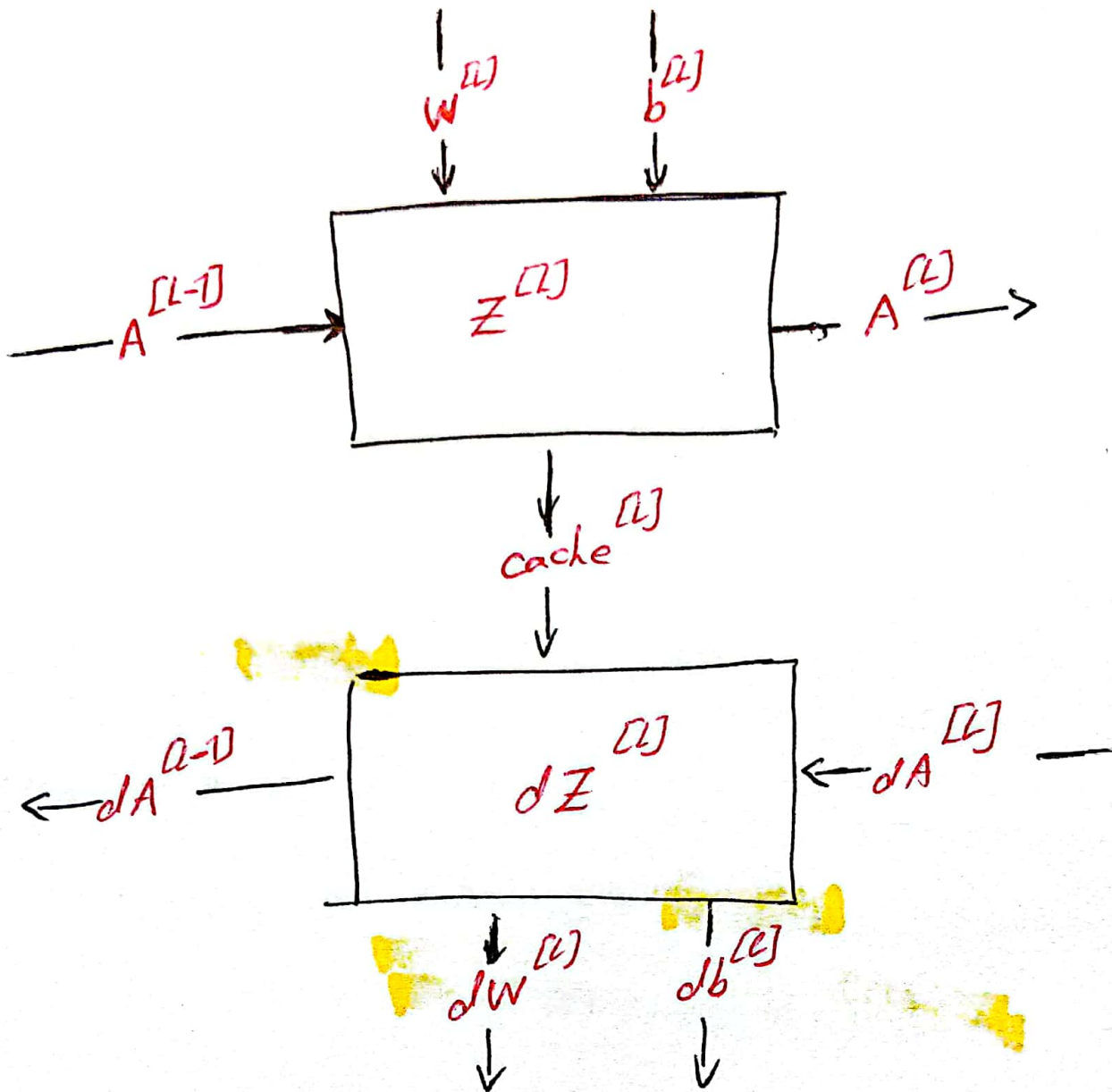
forward propagation is seeded with $A^{[0]} = X$ and evaluates a set of recurrence relations to compute the predictions $A^{[L]} = \hat{Y}$ and then compute the cost: $J = f(\hat{Y}, Y) = f(A^{[L]}, Y)$

backward propagation, on the other hand, is seeded with $\frac{\partial J}{\partial A^{[L]}} = \frac{\partial J}{\partial \hat{Y}}$ and evaluates a different set of recurrence relations to compute $\frac{\partial J}{\partial W^{[L]}}$ and $\frac{\partial J}{\partial b^{[L]}}$.
- if not stopped permanently, it eventually computes

$$\frac{\partial J}{\partial A^{[0]}} = \frac{\partial J}{\partial X}, \text{ a partial derivative we usually ignore.}$$

visualization of forward and Backward propagation.

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Activation function

mathematic behind "Relu" (rectified linear unit)

$$a_{j,i}^{[L]} = g_j^{[L]}(z_{1,i}^{[L]}, \dots, z_{j,i}^{[L]}, \dots, z_{n,i}^{[L]})$$

$$= \max(0, z_{j,i}^{[L]})$$

$$= \begin{cases} z_{ji}^{[l]} & \text{if } z_{ji}^{[l]} > 0, \\ 0 & \text{otherwise,} \end{cases}$$

in other words: $A^{[l]} = \max(0, Z^{[l]})$

Partial derivative of the activations

$$\frac{\partial a_{ji}^{[l]}}{\partial z_{ji}^{[l]}} = \begin{cases} 1 & \text{if } z_{ji}^{[l]} > 0, \\ 0 & \text{otherwise} \end{cases}$$

$$= I(z_{ji}^{[l]} > 0),$$

$$\frac{\partial a_{pi}^{[l]}}{\partial z_{ji}^{[l]}} = 0, \quad \forall p \neq j$$

it follows that:

$$\frac{\partial J}{\partial z_{ji}^{[l]}} = \sum_p \frac{\partial J}{\partial a_{pi}^{[l]}} \cdot \frac{\partial a_{pi}^{[l]}}{\partial z_{ji}^{[l]}} \quad \xrightarrow{\text{next}}$$

$$= \frac{\partial J}{\partial a_{j,i}^{[L]}} \cdot \frac{\partial a_{j,i}^{[L]}}{\partial z_{j,i}^{[L]}} + \sum_{p \neq j} \frac{\partial J}{\partial a_{p,i}^{[L]}} \cdot \frac{\partial a_{p,i}^{[L]}}{\partial z_{j,i}^{[L]}} \quad (12)$$

$$= \frac{\partial J}{\partial a_{j,i}^{[L]}} I(z_{j,i}^{[L]} > 0)$$

Vectorized form:

$$\frac{\partial J}{\partial \mathbf{z}^{[L]}} = \frac{\partial J}{\partial \mathbf{A}^{[L]}} \odot I(\mathbf{z}^{[L]} > 0)$$

⊙ elementwise multiplication.

mathematics behind Sigmoid activation function

$$a_{j,i}^{[L]} = g_j(z_{1,i}^{[L]}, \dots, z_{j,i}^{[L]}, \dots, z_{n,i}^{[L]})$$

$$= \sigma(z_{j,i}^{[L]})$$

$$= \frac{1}{1 + \exp(-z_{j,i}^{[L]})}$$

vectorization of sigmoid activation function

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$$A^{[1]} = \frac{1}{1 + \exp(-Z^{[1]})}$$

To practice backpropagation, first, we construct a computation graph:

$$u_0 = Z_{j,i}^{[1]}$$

$$u_1 = -u_0;$$

$$u_2 = \exp(u_1);$$

$$u_3 = 1 + u_2;$$

$$u_4 = \frac{1}{u_3} = \alpha_{j,i}^{[1]}$$

then we perform an outside first traversal of the

chain rule:

$$\frac{\partial a_{j,i}^{[1]}}{\partial u_4} = 1;$$

$$\frac{\partial a_{j,i}^{[1]}}{\partial u_3} = \frac{\partial a_{j,i}^{[1]}}{\partial u_4} \cdot \frac{\partial u_4}{\partial u_3} = -\frac{1}{u_3^2} = \frac{1}{(1 + \exp(-Z_{j,i}^{[1]}))^2}$$

$$\frac{\partial a_{jii}^{(l)}}{\partial u_2} = \frac{\partial a_{jii}^{(l)}}{\partial u_3} \cdot \frac{\partial u_3}{\partial u_2} = \frac{-1}{u_3^2} = -\frac{1}{(1 + \exp(-z_{jii}^{(l)}))^2};$$

$$\frac{\partial a_{jii}^{(l)}}{\partial u_4} = \frac{\partial a_{jii}^{(l)}}{\partial u_2} \cdot \frac{\partial u_2}{\partial u_4} = -\frac{1}{u_3^2} \exp(u_4) = -\frac{\exp(-z_{jii}^{(l)})}{(1 + \exp(-z_{jii}^{(l)}))^2}$$

$$\frac{\partial a_{jii}^{(l)}}{\partial u_0} = \frac{\partial a_{jii}^{(l)}}{\partial u_1} \cdot \frac{\partial u_1}{\partial u_0} = \frac{1}{u_3^2} \exp(u_4) = \frac{\exp(-z_{jii}^{(l)})}{(1 + \exp(-z_{jii}^{(l)}))^2}$$

Let us simplify:

$$\begin{aligned} \frac{\partial a_{jii}^{(l)}}{\partial z_{jii}^{(l)}} &= \frac{\exp(-z_{jii}^{(l)})}{(1 + \exp(-z_{jii}^{(l)}))^2} \\ &= \frac{1 + \exp(-z_{jii}^{(l)}) - 1}{(1 + \exp(-z_{jii}^{(l)}))^2} \end{aligned}$$

$$= \frac{1}{1 + \exp(-z_{jii}^{(l)})} - \frac{1}{(1 + \exp(-z_{jii}^{(l)}))^2}$$

$$= a_{jii}^{(l)} (1 - a_{jii}^{(l)})$$

we also note that $\frac{\partial a_{pji}^{(l)}}{\partial z_{jii}^{(l)}} = 0, \forall p \neq j$.

consequently:

$$\begin{aligned}\frac{\partial J}{\partial z_{ji}^{(L)}} &= \sum_p \frac{\partial J}{\partial a_{pji}^{(L)}} \cdot \frac{\partial a_{pji}^{(L)}}{\partial z_{ji}^{(L)}} \\&= \frac{\partial J}{\partial a_{jji}^{(L)}} \cdot \frac{\partial a_{jji}^{(L)}}{\partial z_{ji}^{(L)}} + \sum_{p \neq j} \frac{\partial J}{\partial a_{pji}^{(L)}} \cdot \frac{\partial a_{pji}^{(L)}}{\partial z_{ji}^{(L)}} \\&= \frac{\partial J}{\partial a_{jji}^{(L)}} a_{jji}^{(L)} (1 - a_{jji}^{(L)}).\end{aligned}$$

vectorization:

$$\frac{\partial J}{\partial \mathbf{z}^{(L)}} = \frac{\partial J}{\partial \mathbf{A}^{(L)}} \odot \mathbf{A}^{(L)} \odot (\mathbf{1} - \mathbf{A}^{(L)}).$$

mathematic behind Tanh activation function

The hyperbolic tangent function (tanh activation function), is given by

$$a_{jji}^{(L)} = g_j^{(L)}(z_{1ji}^{(L)}, \dots, z_{jji}^{(L)}, \dots, z_{n^{(L)},ji}^{(L)})$$

$$= \tanh(z_{jji}^{(L)})$$

$$= \frac{\exp(z_{jji}^{(L)}) - \exp(-z_{jji}^{(L)})}{\exp(z_{jji}^{(L)}) + \exp(-z_{jji}^{(L)})}$$

By utilizing element-wise multiplication, we get:

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$$A^{(l)} = \frac{1}{\exp(Z^{(l)}) + \exp(-Z^{(l)})} \odot (\exp(Z^{(l)}) - \exp(-Z^{(l)}))$$

Intermediate variables for practicing Back Propagation

$$u_0 = Z_{jii}^{(l)}$$

$$u_1 = -u_0$$

$$u_2 = \exp(u_0)$$

$$u_3 = \exp(u_1)$$

$$u_4 = u_2 - u_3$$

$$u_5 = u_2 + u_3$$

$$u_6 = \frac{1}{u_5}$$

$$Z \odot u_4 u_6 = a_{jii}^{(l)}$$

next, we compute the partial derivatives:

$$\frac{\partial a_{jii}^{(l)}}{\partial u_7} = 1$$

$$\frac{\partial a_{jii}^{(l)}}{\partial u_6} = \frac{\partial a_{jii}^{(l)}}{\partial u_7} \cdot \frac{\partial u_7}{\partial u_6} = u_4 = \exp(Z_{jii}^{(l)}) - \exp(-Z_{jii}^{(l)})$$

$$\frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_5} = \frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_6} \cdot \frac{\partial \mu_6}{\partial \mu_5} = -\mu_4 \frac{1}{\mu_5^2} = -\frac{\exp(\tau_{jji}^{(l)}) - \exp(-\tau_{jji}^{(l)})}{(\exp(\tau_{jji}^{(l)}) + \exp(-\tau_{jji}^{(l)}))^2} \quad (17)$$

$$\frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_u} = \frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_7} \cdot \frac{\partial \mu_7}{\partial \mu_u} = \mu_6 = \frac{1}{\exp(\tau_{jji}^{(l)}) + \exp(-\tau_{jji}^{(l)})};$$

$$\frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_3} = \frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_u} \cdot \frac{\partial \mu_u}{\partial \mu_3} + \frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_5} \cdot \frac{\partial \mu_5}{\partial \mu_3};$$

$$= -\mu_6 - \mu_4 \frac{1}{\mu_5^2}$$

$$= -\frac{1}{\exp(\tau_{jji}^{(l)}) + \exp(-\tau_{jji}^{(l)})} - \frac{\exp(\tau_{jji}^{(l)}) - \exp(-\tau_{jji}^{(l)})}{(\exp(\tau_{jji}^{(l)}) + \exp(-\tau_{jji}^{(l)}))^2}$$

$$= -\frac{2 \exp(\tau_{jji}^{(l)})}{(\exp(\tau_{jji}^{(l)}) + \exp(-\tau_{jji}^{(l)}))^2};$$

$$\frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_2} = \frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_u} \cdot \frac{\partial \mu_u}{\partial \mu_2} + \frac{\partial \alpha_{jji}^{(l)}}{\partial \mu_5} \cdot \frac{\partial \mu_5}{\partial \mu_2};$$

$$= \mu_6 - \mu_4 \frac{1}{\mu_5^2}$$

$$= \frac{1}{\exp(z_{jji}^{(L)}) + \exp(-z_{jji}^{(L)})} - \frac{\exp(z_{jji}^{(L)}) - \exp(-z_{jji}^{(L)})}{(\exp(z_{jji}^{(L)}) + \exp(-z_{jji}^{(L)}))^2}; \quad (12)$$

$$= \frac{2 \exp(-z_{jji}^{(L)})}{(\exp(z_{jji}^{(L)}) + \exp(-z_{jji}^{(L)}))^2};$$

$$\frac{\partial a_{jji}^{(L)}}{\partial \mu_1} = \frac{\partial a_{jji}^{(L)}}{\partial \mu_3} \cdot \frac{\partial \mu_3}{\partial \mu_1};$$

$$= (-\mu_6 - \mu_4 \frac{1}{\mu_5^2}) \exp(\mu_1);$$

$$= - \frac{2 \exp(z_{jji}^{(L)}) \exp(-z_{jji}^{(L)})}{(\exp(z_{jji}^{(L)}) + \exp(-z_{jji}^{(L)}))^2};$$

$$\frac{\partial a_{jji}^{(L)}}{\partial \mu_0} = \frac{\partial a_{jji}^{(L)}}{\partial \mu_1} \cdot \frac{\partial \mu_1}{\partial \mu_0} + \frac{\partial a_{jji}^{(L)}}{\partial \mu_2} \cdot \frac{\partial \mu_2}{\partial \mu_0}$$

$$= -(-\mu_6 - \mu_4 \frac{1}{\mu_5^2}) \exp(\mu_1) + (\mu_6 - \mu_4 \frac{1}{\mu_5^2}) \exp(\mu_0)$$

$$= \frac{2 \exp(z_{jji}^{(L)}) \exp(-z_{jji}^{(L)})}{(\exp(z_{jji}^{(L)}) + \exp(-z_{jji}^{(L)}))^2} + \frac{2 \exp(z_{jji}^{(L)}) \exp(-z_{jji}^{(L)})}{(\exp(z_{jji}^{(L)}) + \exp(-z_{jji}^{(L)}))^2};$$

$$= \frac{4 \exp(z_{jji}^{(l)}) \exp(-z_{jji}^{(l)})}{(\exp(z_{jji}^{(l)}) + \exp(-z_{jji}^{(l)}))^2}$$

it follows that:

$$\begin{aligned} \frac{\partial a_{jji}^{(l)}}{\partial z_{jji}^{(l)}} &= \frac{4 \exp(z_{jji}^{(l)}) \exp(-z_{jji}^{(l)})}{(\exp(z_{jji}^{(l)}) + \exp(-z_{jji}^{(l)}))^2} \\ &= \frac{\exp(z_{jji}^{(l)})^2 + 2 \exp(z_{jji}^{(l)}) \cdot \exp(-z_{jji}^{(l)}) + \exp(-z_{jji}^{(l)})^2}{(\exp(z_{jji}^{(l)}) + \exp(-z_{jji}^{(l)}))^2} \\ &\quad - \frac{\exp(z_{jji}^{(l)})^2 - 2 \exp(z_{jji}^{(l)}) \exp(-z_{jji}^{(l)}) + \exp(-z_{jji}^{(l)})^2}{(\exp(z_{jji}^{(l)}) + \exp(-z_{jji}^{(l)}))^2} \end{aligned}$$

$$= 1 - \frac{(\exp(z_{jji}^{(l)}) - \exp(-z_{jji}^{(l)}))^2}{(\exp(z_{jji}^{(l)}) + \exp(-z_{jji}^{(l)}))^2}$$

$$= 1 - \sigma_{jji}^{(l)} \cdot a_{jji}^{(l)}$$

we also have:

$$\frac{\partial a_{pji}^{(l)}}{\partial z_{jji}^{(l)}} = 0, \quad \forall p \neq j$$

Thus:

$$\begin{aligned}
 \frac{\partial J}{\partial z_{ji}^{(L)}} &= \sum_p \frac{\partial J}{\partial o_{ji}^{(L)}} \cdot \frac{\partial a_{pi}^{(L)}}{\partial z_{ji}^{(L)}} \\
 &= \frac{\partial J}{\partial a_{ji}^{(L)}} \cdot \frac{\partial a_{ji}^{(L)}}{\partial z_{ji}^{(L)}} + \sum_p \frac{\partial J}{\partial a_{ji}^{(L)}} \cdot \frac{\partial a_{pi}^{(L)}}{\partial z_{ji}^{(L)}} \\
 &= \frac{\partial J}{\partial a_{ji}^{(L)}} (1 - a_{ji}^{(L)} \cdot a_{ji}^{(L)}) ;
 \end{aligned}$$

which implies that

$$\frac{\partial J}{\partial z^{(L)}} = \frac{\partial J}{\partial A^{(L)}} \odot (1 - A^{(L)} \odot A^{(L)})$$

Mathematics behind soft max activation function

$$\begin{aligned}
 a_{ji}^{(L)} &= g_j^{(L)}(z_{1,i}^{(L)}, \dots, z_{ji}^{(L)}, \dots, z_{n,i}^{(L)}) \\
 &= \frac{\exp(z_{ji}^{(L)})}{\sum_p \exp(z_{pi}^{(L)})}
 \end{aligned}$$

vectorization results:

$$A^{[l]} = \frac{1}{\text{broadcast}\left(\underbrace{\sum_{\text{axis}=0} \exp(Z^{[l]})}_{\text{row vector}}\right)} \odot \exp(Z^{[l]}).$$

To begin with, we construct a computation graph for the j th activation of the current layer:

$$\mu_{-1} = Z_{j,i}^{[l]};$$

$$\mu_{0,p} = Z_{p,i}^{[l]}; \quad \forall p \neq j;$$

$$\mu_1 = \exp(\mu_{-1});$$

$$\mu_{2,p} = \exp(\mu_{0,p}), \quad \forall p \neq j;$$

$$\mu_3 = \mu_1 + \sum_{p \neq j} \mu_{2,p};$$

$$\mu_4 = \frac{1}{\mu_3};$$

$$\mu_5 = \mu_1 \cdot \mu_4 = a_{j,i}^{[l]};$$

by applying the chain rule, we get:

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$$\frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_5} = 1;$$

$$\frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_4} = \frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_5} \cdot \frac{\partial \mu_5}{\partial \mu_4} = \mu_1 = \exp(\mathbb{Z}_{jii}^{[1]});$$

$$\frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_3} = \frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_4} \cdot \frac{\partial \mu_4}{\partial \mu_3} = -\mu_1 \frac{1}{\mu_3^2} = -\frac{\exp(\mathbb{Z}_{jii}^{[1]})}{(\sum_p \exp(\mathbb{Z}_{p,ii}^{[1]}))^2};$$

$$\frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_1} = \frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_3} \cdot \frac{\partial \mu_3}{\partial \mu_1} + \frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_5} \cdot \frac{\partial \mu_5}{\partial \mu_1}$$

$$= -\mu_1 \frac{1}{\mu_3^2} + \mu_4;$$

$$= -\frac{\exp(\mathbb{Z}_{jii}^{[1]})}{(\sum_p \exp(\mathbb{Z}_{p,ii}^{[1]}))^2} + \frac{1}{\sum_p \exp(\mathbb{Z}_{p,ii}^{[1]})};$$

$$\frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_{-1}} = \frac{\partial \alpha_{jii}^{[1]}}{\partial \mu_1} \cdot \frac{\partial \mu_1}{\partial \mu_{-1}} = \left(-\mu_1 \frac{1}{\mu_3^2} + \mu_4\right) \exp(\mu_{-1})$$

$$= \frac{\exp(\mathbb{Z}_{jii}^{[1]})^2}{(\sum_p \exp(\mathbb{Z}_{p,ii}^{[1]}))^2} + \frac{\exp(\mathbb{Z}_{jii}^{[1]})}{\sum_p \exp(\mathbb{Z}_{p,ii}^{[1]})}.$$

next, we need to take into account that $z_{ji}^{(L)}$ also (23)
affects other activations in the same layer,

$$u_1 = z_{ji}^{(L)};$$

$$u_{0,p} = z_{p,i}^{(L)}, \quad \forall p \neq j;$$

$$u_1 = \exp(u_1);$$

$$u_{2,p} = \exp(u_{0,p}), \quad \forall p \neq j;$$

$$u_3 = u_1 + \sum_{p \neq j} u_{2,p};$$

$$u_4 = \frac{1}{u_3};$$

$$u_5 = u_{2,p} \cdot u_4 = a_{p,i}^{(L)}, \quad \forall p \neq j;$$

Backward propagation gives us the remaining partial

derivatives:

$$\frac{\partial a_{p,i}^{(L)}}{\partial u_5} = 1;$$

$$\frac{\partial a_{p,i}^{(L)}}{\partial u_4} = \frac{\partial a_{p,i}^{(L)}}{\partial u_5} \cdot \frac{\partial u_5}{\partial u_4} = u_{2,p} = \exp(z_{p,i}^{(L)});$$

$$\frac{\partial a_{pi}^{(L)}}{\partial u_3} = \frac{\partial a_{pi}^{(L)}}{\partial u_4} \cdot \frac{\partial u_4}{\partial u_3} = -u_{2,p} \frac{1}{u_3^2} = - \frac{\exp(Z_{pi}^{(L)})}{(\sum_p \exp(Z_{pi}^{(L)}))^2}; \quad (24)$$

$$\frac{\partial a_{pi}^{(L)}}{\partial u_1} = \frac{\partial a_{pi}^{(L)}}{\partial u_3} \cdot \frac{\partial u_3}{\partial u_1} = -u_{2,p} \frac{1}{u_3^2} = - \frac{\exp(Z_{pi}^{(L)})}{(\sum_p \exp(Z_{pi}^{(L)}))^2}$$

$$\begin{aligned} \frac{\partial a_{pi}^{(L)}}{\partial a_{-1}} &= \frac{\partial a_{pi}^{(L)}}{\partial u_1} \cdot \frac{\partial u_1}{\partial a_{-1}} = -u_{2,p} \frac{1}{u_3^2} \exp(u-1) \\ &= - \frac{\exp(Z_{pi}^{(L)}) \exp(Z_{ji}^{(L)})}{(\sum_p \exp(Z_{pi}^{(L)}))^2} \end{aligned}$$

we now know that:

$$\begin{aligned} \frac{\partial a_{ji}^{(L)}}{\partial Z_{ji}^{(L)}} &= - \frac{\exp(Z_{ji}^{(L)})^2}{(\sum_p \exp(Z_{pi}^{(L)}))^2} + \frac{\exp(Z_{ji}^{(L)})}{\sum_p \exp(Z_{pi}^{(L)})} \\ &= a_{ji}^{(L)} (1 - a_{ji}^{(L)}); \end{aligned}$$

$$\frac{\partial a_{pi}^{(L)}}{\partial Z_{ji}^{(L)}} = - \frac{\exp(Z_{pi}^{(L)}) \exp(Z_{ji}^{(L)})}{(\sum_p \exp(Z_{pi}^{(L)}))^2} = -a_{pi}^{(L)} \cdot a_{ji}^{(L)}; \quad \forall p \neq j.$$

hence:

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$$\begin{aligned}
 \frac{\partial J}{\partial z_{ji}^{(L)}} &= \sum_p \frac{\partial J}{\partial a_{pi}^{(L)}} \cdot \frac{\partial a_{pi}^{(L)}}{\partial z_{ji}^{(L)}} \\
 &= \frac{\partial J}{\partial a_{ji}^{(L)}} \cdot \frac{\partial a_{ji}^{(L)}}{\partial z_{ji}^{(L)}} + \sum_{p \neq j} \frac{\partial J}{\partial a_{pi}^{(L)}} \cdot \frac{\partial a_{pi}^{(L)}}{\partial z_{ji}^{(L)}} \\
 &= \frac{\partial J}{\partial a_{ji}^{(L)}} a_{ji}^{(L)} (1 - a_{ji}^{(L)}) - \sum_{p \neq j} \frac{\partial J}{\partial a_{pi}^{(L)}} a_{pi}^{(L)} \cdot a_{ji}^{(L)} \\
 &= a_{ji}^{(L)} \left(\frac{\partial J}{\partial a_{ji}^{(L)}} (1 - a_{ji}^{(L)}) - \sum_{p \neq j} \frac{\partial J}{\partial a_{pi}^{(L)}} a_{pi}^{(L)} + \frac{\partial J}{\partial a_{ji}^{(L)}} a_{ji}^{(L)} \right) \\
 &= a_{ji}^{(L)} \left(\frac{\partial J}{\partial a_{ji}^{(L)}} - \sum_p \frac{\partial J}{\partial a_{pi}^{(L)}} a_{pi}^{(L)} \right);
 \end{aligned}$$

which we can vectorize as:

$$\frac{\partial J}{\partial z_{:ji}^{(L)}} = a_{:ji}^{(L)} \odot \left(\frac{\partial J}{\partial a_{:ji}^{(L)}} - \underbrace{a_{:ji}^{(L)T} \frac{\partial J}{\partial a_{:ji}^{(L)}}}_{\text{scalar}} \right)$$

↓

$$\frac{\partial J}{\partial z^{(L)}} = A^{(L)} \odot \left(\frac{\partial J}{\partial A^{(L)}} - \text{broadcast} \left(\sum_{\text{axis} \neq 0} \frac{\partial J}{\partial A^{(L)}} \odot A^{(L)} \right) \right)$$

mathematic behind cost functions

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Binary classification

the cost function of binary classification

$$J = f(\hat{Y}, Y) = f(A^{[L]}, Y)$$

$$= -\frac{1}{m} \sum_i (y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i))$$

$$= -\frac{1}{m} \sum_i (y_i \log(a_i^{[L]}) + (1-y_i) \log(1-a_i^{[L]}))$$

which we can write as

$$J = -\frac{1}{m} \sum_{\text{over } i=1} (Y \odot \log(A^{[L]}) + (1-Y) \odot \log(1-A^{[L]}))$$

Scalar

Next, we construct a computational graph:

$$u_{0,i} = a_i^{[L]};$$

$$u_{1,i} = 1 - u_{0,i};$$

$$u_{2,i} = \log(u_{0,i});$$

$$u_{3,i} = \log(u_{1,i});$$

$$u_{4,i} = y_i u_{2,i} + (1-y_i) u_{3,i}$$
$$u_5 = -\frac{1}{m} \sum_i u_{4,i} = J$$

Derivative computations are now as simple as they get, 27

$$\frac{\partial J}{\partial \mu_5} = 1;$$

$$\frac{\partial J}{\partial \mu_{u,i}} = \frac{\partial J}{\partial \mu_5} \cdot \frac{\partial \mu_5}{\partial \mu_{u,i}} = -\frac{1}{m};$$

$$\frac{\partial J}{\partial \mu_{3,i}} = \frac{\partial J}{\partial \mu_{u,i}} \cdot \frac{\partial \mu_{u,i}}{\partial \mu_{3,i}} = -\frac{1}{m} (1 - y_i);$$

$$\frac{\partial J}{\partial \mu_{2,i}} = \frac{\partial J}{\partial \mu_{u,i}} \cdot \frac{\partial \mu_{u,i}}{\partial \mu_{2,i}} = -\frac{1}{m} y_i;$$

$$\frac{\partial J}{\partial \mu_{4,i}} = \frac{\partial J}{\partial \mu_{3,i}} \cdot \frac{\partial \mu_{3,i}}{\partial \mu_{4,i}} = -\frac{1}{m} (1 - y_i) \frac{1}{\mu_{1,i}} = -\frac{1}{m} \cdot \frac{1 - y_i}{1 - \alpha_i^{(L)}};$$

$$\begin{aligned} \frac{\partial J}{\partial \mu_{0,i}} &= \frac{\partial J}{\partial \mu_{1,i}} \cdot \frac{\partial \mu_{1,i}}{\partial \mu_{0,i}} + \frac{\partial J}{\partial \mu_{2,i}} \cdot \frac{\partial \mu_{2,i}}{\partial \mu_{0,i}} = \frac{1}{m} (1 - y_i) \frac{1}{\mu_{1,i}} - \frac{1}{m} y_i \frac{1}{\mu_{0,i}} \\ &= \frac{1}{m} \left(\frac{1 - y_i}{1 - \alpha_i^{(L)}} - \frac{y_i}{\alpha_i^{(L)}} \right) \end{aligned}$$

thus: $\frac{\partial J}{\partial \alpha_i^{(L)}} = \frac{1}{m} \left(\frac{1 - y_i}{1 - \alpha_i^{(L)}} - \frac{y_i}{\alpha_i^{(L)}} \right);$

which implies that: $\frac{\partial J}{\partial A^{(L)}} = \frac{1}{m} \left(\frac{1}{1 - A^{(L)}} \odot (1 - Y) - \frac{1}{A^{(L)}} \odot Y \right).$

In addition, since the sigmoid activation function is used 28

in the output layer, we get:

$$\begin{aligned}\frac{\partial J}{\partial I_i^{[L]}} &= \frac{\partial J}{\partial a_i^{[L]}} \cdot a_i^{[L]} (1 - a_i^{[L]}) \\ &= \frac{1}{m} \left(\frac{1 - y_i}{1 - a_i^{[L]}} - \frac{y_i}{a_i^{[L]}} \right) a_i^{[L]} (1 - a_i^{[L]}); \\ &= \frac{1}{m} ((1 - y_i) a_i^{[L]} - y_i (1 - a_i^{[L]})) \\ &= \frac{1}{m} (a_i^{[L]} - y_i); \end{aligned}$$

in other words:

$$\frac{\partial J}{\partial I^{[L]}} = \frac{1}{m} (A^{[L]} - Y).$$

note that both $\frac{\partial J}{\partial I^{[L]}} \in \mathbb{R}^{1 \times m}$ and $\frac{\partial J}{\partial A^{[L]}} \in \mathbb{R}^{1 \times m}$,
because $n^{[L]} = 1$ in this case.

multiclass classification

in multiclass classification, the cost function is given by:

$$J = f(\hat{Y}, Y) = f(A^{[L]}, Y);$$

$$= -\frac{1}{m} \sum_i \sum_j y_{ji} \log(\hat{y}_{ji});$$

$$= -\frac{1}{m} \sum_i \sum_j y_{ji} \log(a_{ji}^{[L]});$$

where $j = 1, \dots, n^{[L]}$

we can vectorized the cost expression:

$$J = -\frac{1}{m} \sum_{\substack{\text{axis}=0 \\ \text{axis}=1}} Y \odot \log(A^{[L]})$$

scalar

next, let us introduce intermediate variables:

$$\mu_{0,ji} = a_{ji}^{[L]};$$

$$\mu_{1,ji} = \log(\mu_{0,ji});$$

$$\mu_{2,ji} = y_{ji} \mu_{1,ji};$$

$$\mu_{3,i} = \sum_j \mu_{2,ji}$$

$$\mu_4 = -\frac{1}{m} \sum_i \mu_{3,i} = J$$

with the computational graph in place, we can
computation

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perform backward propagation:

$$\frac{\partial J}{\partial \mu_4} = 1 ;$$

$$\frac{\partial J}{\partial \mu_{3,i}} = \frac{\partial J}{\partial \mu_4} \cdot \frac{\partial \mu_4}{\partial \mu_{3,i}} = -\frac{1}{m} ;$$

$$\frac{\partial J}{\partial \mu_{2,j,i}} = \frac{\partial J}{\partial \mu_{3,i}} \cdot \frac{\partial \mu_{3,i}}{\partial \mu_{2,j,i}} = -\frac{1}{m} y_{j,i} ;$$

$$\frac{\partial J}{\partial \mu_{0,j,i}} = \frac{\partial J}{\partial \mu_{2,j,i}} \cdot \frac{\partial \mu_{2,j,i}}{\partial \mu_{0,j,i}} = -\frac{1}{m} y_{j,i} \frac{1}{\mu_{0,j,i}} = -\frac{1}{m} \frac{y_{j,i}}{\alpha_{j,i}^{[L]}} ;$$

hence:

$$\frac{\partial J}{\partial \alpha_{j,i}^{[L]}} = -\frac{1}{m} \frac{y_{j,i}}{\alpha_{j,i}^{[L]}} ;$$

vectorization is trivial:

$$\frac{\partial J}{\partial A^{[L]}} = -\frac{1}{m} \cdot \frac{1}{A^{[L]}} \odot Y ;$$

furthermore, since the output layer uses the
soft max activation function, we get:

$$\frac{\partial J}{\partial I_{jii}^{[L]}} = a_{jii}^{[L]} \left(\frac{\partial J}{\partial a_{jii}^{[L]}} - \sum_p \frac{\partial J}{\partial a_{p,ii}^{[L]}} a_{p,ii}^{[L]} \right) \quad (37)$$

$$= a_{jii}^{[L]} \left(-\frac{1}{m} \cdot \frac{y_{jii}}{a_{jii}^{[L]}} + \sum_p \frac{1}{m} \cdot \frac{y_{p,ii}}{a_{p,ii}^{[L]}} a_{p,ii}^{[L]} \right)$$

$$= \frac{1}{m} \left(-y_{jii} + a_{jii}^{[L]} \underbrace{\sum_p y_{p,ii}}_{\sum \text{probabilities} = 1} \right)$$

$$= \frac{1}{m} (a_{jii}^{[L]} - y_{jii})$$

note that $p = 1, \dots, n^{[L]}$

to conclude:

$$\frac{\partial J}{\partial I^{[L]}} = \frac{1}{m} (A^{[L]} - Y).$$

multi label classification

we can view multi-label classification as j binary classification problems:

$$J = f(\hat{Y}, Y) = f(A^{[L]}, Y);$$

$$= \sum_j \left(-\frac{1}{m} \sum_i (y_{ji} \log(\hat{y}_{ji}) + (1 - y_{ji}) \log(1 - \hat{y}_{ji})) \right);$$

$$= \sum_j \left(-\frac{1}{m} \sum_i (y_{ji} \log(a_{ji}^{[L]}) + (1 - y_{ji}) \log(1 - a_{ji}^{[L]})) \right);$$

where once again $j = 1, \dots, n^{[L]}$.

Vectorization gives:

$$J = -\frac{1}{m} \sum_{\substack{\text{axis}=1 \\ \text{axis}=0}} (Y \odot \log(A^{[L]}) + (1 - Y) \odot \log(1 - A^{[L]}))$$

scalar

the computation graph:

$$\mu_{0,ji} = a_{ji}^{[L]}$$

$$\mu_{1,ji} = 1 - \mu_{0,ji}$$

$$\mu_{2,ji} = \log(\mu_{0,ji})$$

$$\mu_{3,ji} = \log(\mu_{1,ji})$$

$$\mu_{4,ji} = y_{ji} \mu_{2,ji} + (1 - y_{ji}) \mu_{3,ji}$$

$$\mu_{5,j} = -\frac{1}{m} \sum_i \mu_{4,ji} \rightarrow \mu_6 = \sum_j \mu_{5,j} = J$$

next, we compute the partial derivatives:

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$$\frac{\partial J}{\partial \mu_0} = 1;$$

$$\frac{\partial J}{\partial \mu_{5ji}} = \frac{\partial J}{\partial \mu_0} \cdot \frac{\partial \mu_0}{\partial \mu_{5ji}} = 1;$$

$$\frac{\partial J}{\partial \mu_{4ji}} = \frac{\partial J}{\partial \mu_{5ji}} \cdot \frac{\partial \mu_{5ji}}{\partial \mu_{4ji}} = -\frac{1}{m};$$

$$\frac{\partial J}{\partial \mu_{3ji}} = \frac{\partial J}{\partial \mu_{4ji}} \cdot \frac{\partial \mu_{4ji}}{\partial \mu_{3ji}} = -\frac{1}{m} (1 - y_{ji});$$

$$\frac{\partial J}{\partial \mu_{2ji}} = \frac{\partial J}{\partial \mu_{3ji}} \cdot \frac{\partial \mu_{3ji}}{\partial \mu_{2ji}} = -\frac{1}{m} y_{ji};$$

$$\frac{\partial J}{\partial \mu_{1ji}} = \frac{\partial J}{\partial \mu_{2ji}} \cdot \frac{\partial \mu_{2ji}}{\partial \mu_{1ji}} = -\frac{1}{m} (1 - y_{ji}) \frac{1}{\mu_{1ji}} = -\frac{1}{m} \cdot \frac{1 - y_{ji}}{1 - \alpha_{ji}^{LL}};$$

$$\frac{\partial J}{\partial \mu_{0ji}} = \frac{\partial J}{\partial \mu_{1ji}} \cdot \frac{\partial \mu_{1ji}}{\partial \mu_{0ji}} + \frac{\partial J}{\partial \mu_{2ji}} \cdot \frac{\partial \mu_{2ji}}{\partial \mu_{0ji}}$$

$$= -\frac{1}{m} (1 - y_{ji}) \frac{1}{\mu_{1ji}} - \frac{1}{m} y_{ji} \frac{1}{\mu_{0ji}}$$

$$= -\frac{1}{m} \left(\frac{1 - y_{ji}}{1 - \alpha_{ji}^{LL}} - \frac{y_{ji}}{\alpha_{ji}^{LL}} \right).$$